

Geometry (10 th std)

Choose the Appropriate Answer

1.If in triangles ABC and EDF, ABDE = BCDF then they will be similar, when

- (a) $\angle B = \angle E$ (b) $\angle A = \angle D$ (c) $\angle B = \angle D$ (d) $\angle A = \angle F$

Ans: (c) $\angle B = \angle D$

2.In $\triangle LMN$, $\angle L = 60^\circ$, $\angle M = 50^\circ$. If $\triangle LMN \sim \triangle PQR$ then the value of $\angle R$ is

- (a) 40° (b) 70° (c) 30° (d) 110°

Ans: (b) 70°

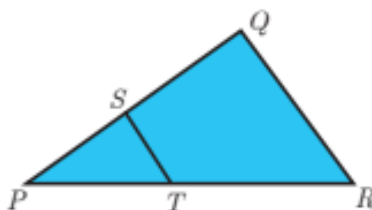
Since $\triangle LMN \sim \triangle PQR$ ($\angle N = \angle R$) $\angle N = 180 - (60 + 50) = 180 - 110^\circ \Rightarrow \angle N = 70^\circ \therefore \angle R = 70^\circ$

3.If $\triangle ABC$ is an isosceles triangle with $\angle C = 90^\circ$ and $AC = 5$ cm, then AB is

- (a) 2.5 cm (b) 5 cm (c) 10 cm (d) $5\sqrt{2}$ cm

Ans: (d) $5\sqrt{2}$ cm

4.In a given figure $ST \parallel QR$, $PS = 2$ cm and $SQ = 3$ cm. Then the ratio of the area of $\triangle PQR$ to the area of $\triangle PST$ is



- (a) 25: 4 (b) 25: 7 (c) 25: 11 (d) 25: 13

Ans: (a) 25: 4

Area of $\triangle PQR$: Area of $\triangle PST = PQ^2: PS^2 = 5^2: 2^2 = 25: 4$

5.The perimeters of two similar triangles $\triangle ABC$ and $\triangle PQR$ are 36 cm and 24 cm respectively. If $PQ = 10$ cm, then the length of AB is

- (a) 6.23 cm (b) $106\sqrt{3}$ (c) 66.23 cm (d) 15 cm

Ans: (d) 15 cm

6.If in $\triangle ABC$, $DE \parallel BC$ (c) $AB = 3.6$ cm, $AC = 2.4$ cm and $AD = 2.1$ cm then the length of AE is

- (a) 1.4 cm (b) 1.8 cm (c) 1.2 cm (d) 1.05 cm

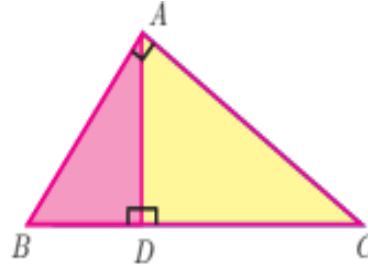
Ans: (a) 1.4 cm

7.In a $\triangle ABC$, AD is the bisector of $\angle BAC$ (c) If $AB = 8$ cm, $BD = 6$ cm and $DC = 3$ cm. The length of the side AC is

- (a) 6 cm (b) 4 cm (c) 3 cm (d) 8 cm

Ans: (b) 4 cm

8. In the adjacent figure $\angle BAC = 90^\circ$ and $AD \perp BC$ then



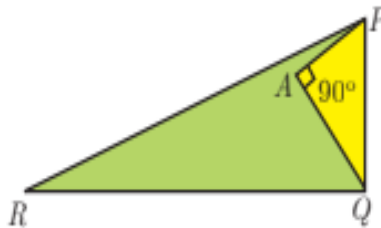
- (a) $BD \cdot CD = BC^2$ (b) $AB \cdot AC = BC^2$ (c) $BD \cdot CD = AD^2$ (d) $AB \cdot AC = AD^2$
Ans: (c) $BD \cdot CD = AD^2$

9. Two poles of heights 6 m and 11 m stand vertically on a plane ground. If the distance between their feet is 12 m, what is the distance between their tops?
 (a) 13 m (b) 14 m (c) 15 m (d) 12.8 m

Ans: (a) 13 m

$$AC^2 = AE^2 + EC^2 = 5^2 + 12^2 = 25 + 144 = 169 \Rightarrow AC = \sqrt{169} = 13 \text{ cm}$$

10. In the given figure, $PR = 26$ cm, $QR = 24$ cm, $\angle PAQ = 90^\circ$, $PA = 6$ cm and $QA = 8$ cm. Find $\angle PQR$.



- (a) 80° (b) 85° (c) 75° (d) 90°
Ans: (d) 90°

11. A tangent is perpendicular to the radius at the

- (a) Centre (b) Point of contact (c) Infinity (d) Chord

Ans: (b) Point of contact

12. How many tangents can be drawn to the circle from an exterior point?

- (a) One (b) Two (c) Infinite (d) Zero

Ans: (b) Two

13. The two tangents from an external point P to a circle with center at O are PA and PB. If $\angle APB = 70^\circ$ then the value of $\angle AOB$ is

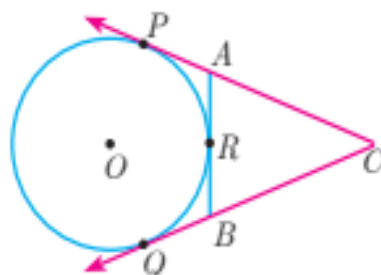
- (a) 100° (b) 110° (c) 120° (d) 130°

Ans: (b) 110°

$$\angle OAP = 90^\circ \Rightarrow \angle APO = 35^\circ$$

$$\angle AOP = 180 - (90 + 35) = 180 - 125 = 55 \Rightarrow \angle AOB = 2 \times 55 = 110^\circ$$

14. In figure CP and CQ are tangents to a circle with center at O. ARB is another tangent touching the circle at R. If CP = 11 cm and BC = 7 cm, then the length of BR is

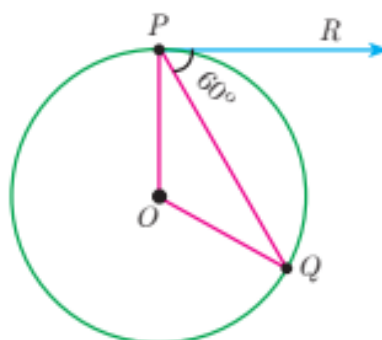


- (a) 6 cm (b) 5 cm (c) 8 cm (d) 4 cm

Ans: (d) 4 cm

$BQ = BR = 4 \text{ cm}$, $PC = QC = 11 \text{ cm}$ (tangent of the circle) $QB + BC = 11 \Rightarrow QB + 7 = 11$ $QB = 11 - 7 = 4 \text{ cm} \Rightarrow BR = BQ = 4 \text{ cm}$

15. In figure if PR is tangent to the circle at P and O is the center of the circle, then $\angle POQ$ is



- (a) 120° (b) 100° (c) 110° (d) 90°

Ans: (a) 120°

$\angle QPR = 90^\circ \Rightarrow \angle OPQ = 90^\circ - 60^\circ = 30^\circ \Rightarrow \angle OQB = 30^\circ$ In $\triangle OPQ \Rightarrow \angle P + \angle Q + \angle O = 180^\circ$ $30 + 30 + \angle O = 180^\circ \Rightarrow \angle O = 180^\circ - 60^\circ = 120^\circ$
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16. If in triangle ABC and DEF, $AB / DE = BC / FD$, then they will be similar when

- (a) $\angle B, \angle E$ (b) $\angle A, \angle D$ (c) $\angle B, \angle D$ (d) $\angle A, \angle F$

Ans: (c) $\angle B, \angle D$

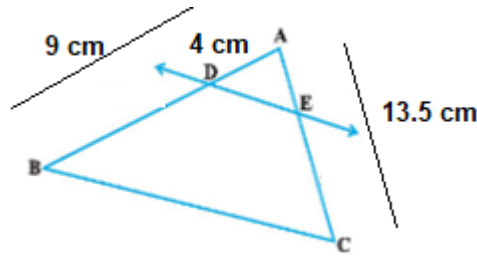
17. It is given that $\triangle ABC \sim \triangle PQR$ with $BC / QR = 1 / 3$, then $\text{ar}(\triangle ABC) / \text{ar}(\triangle PQR)$ is equal to

- (a) 9 (b) 3 (c) $1 / 3$ (d) $1 / 9$

Ans: (d) $1 / 9$

$\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PQR)} = \left(\frac{BC}{QR} \right)^2 = \left(\frac{1}{3} \right)^2 = \frac{1^2}{3^2} = \frac{1}{9}$

18. In $\triangle ABC$, $DE \parallel BC$ and $AD = 4\text{ cm}$, $AB = 9\text{ cm}$. $AC = 13.5\text{ cm}$ then the value of EC is



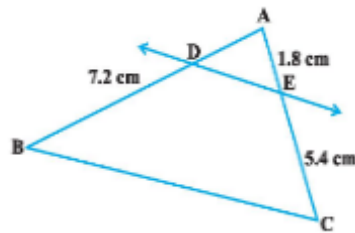
- (a) 6 cm (b) 7.5 cm (c) 9 cm (d) None of these

Ans: (b) 7.5 cm

$$DB = AB - AD = 9\text{ cm} - 4\text{ cm} = 5\text{ cm}$$

$$\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{4}{5} = \frac{13.5 - x}{x} \Rightarrow x = \frac{67.5}{9} = 7.5\text{ cm}$$

19. In figure $DE \parallel BC$ then the value of AD is



- (a) 2 cm (b) 2.4 cm (c) 3 cm (d) None of the above

Ans: (b) 2.4 cm

$$DB = AB - AD = 9\text{ cm} - 4\text{ cm} = 5\text{ cm}$$

$$\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow AD = \frac{AE \times DB}{EC} = \frac{1.8 \times 7.2}{5.4} = 2.4\text{ cm}$$

20. A ladder is placed against a wall such that its foot is at distance of 2.5 m from the wall and its top reaches a window 6 m above the ground. (d) The length of the ladder is

- (a) 6.5 m (b) 7.5 m (c) 8.5 m (d) 9.5 m

Ans: (a) 6.5 m

21. If the corresponding sides of two similar triangles are in the ratio 4 : 9, then the areas of these triangles are in the ratio is

- (a) 2 : 3 (b) 3 : 2 (c) 81 : 16 (d) 16 : 81

Ans: (d) 16: 81

22. If $\triangle ABC \sim \triangle PQR$, $BC = 8\text{ cm}$ and $QR = 6\text{ cm}$, the ratio of the areas of $\triangle ABC$ and $\triangle PQR$ is

- (a) 8 : 6 (b) 6 : 8 (c) 64 : 36 (d) 16 : 9

Ans: (d) 16: 9

23. If $\triangle ABC \sim \triangle PQR$, area of $\triangle ABC = 81\text{cm}^2$, area of $\triangle PQR = 144\text{cm}^2$ and $QR = 6$ cm, then length of BC is
(a) 4 cm (b) 4.5 cm (c) 9 cm (d) 12 cm

Ans: (b) 4.5 cm

24. Sides of triangles are given below. Which of these is a right triangle?
(a) 7 cm, 5 cm, 24 cm (b) 34 cm, 30 cm, 16 cm
(c) 4 cm, 3 cm, 7 cm (d) 8 cm, 12 cm, 14 cm

Ans: (b) 34 cm, 30 cm, 16 cm

25. If a ladder 10 m long reaches a window 8 m above the ground, then the distance of the foot of the ladder from the base of the wall is
(a) 18 m (b) 8 m (c) 6 m (d) 4 m

Ans: (c) 6 m

26. A girl walks 200 towards East and she walks 150m towards North. The distance of the girl from the starting point is
(a) 350m (b) 250m (c) 300m (d) 225m

Ans: (b) 250m

27. All _____ triangles are similar.
(a) Isosceles (b) Equilateral (c) Scalene (d) Right angled

Ans: (b) Equilateral

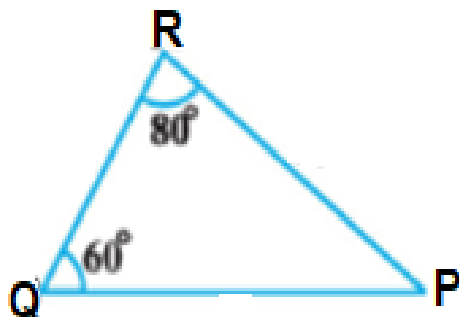
28. All circles are _____
(a) Congruent (b) Similar (c) Not similar (d) None of these

Ans: (b) Similar

29. All squares are _____
(a) Congruent (b) Similar (c) Not similar (d) None of these

Ans: (b) Similar

30. In the given below figure, the value of $\angle P$ is



- (a) 60° (b) 80° (c) 40° (d) 100°

Ans: (c) 40°

31. A girl of height 90 cm is walking away from the base of a lamp-post at a speed of 1.2 m/s. If the lamp is 3.6 m above the ground, then the length of her shadow after 4 seconds.

- (a) 1.2 m (b) 1.6 m (c) 2 m (d) None of these

Ans: (b) 1.6 m

$$\frac{BE}{DE} = \frac{AB}{CD} \Rightarrow \frac{4.8 + x}{x} = \frac{3.6}{0.9} \Rightarrow x = 1.6$$

32. A vertical pole of length 6 m casts a shadow 4 m long on the ground and at the same time a tower casts a shadow 28 m long. Find the height of the tower.

- (a) 42 m (b) 48 m (c) 54 m (d) None of these

Ans: (a) 42 m

$$\frac{\text{Height}_{\text{pole}}}{\text{Shadow}_{\text{pole}}} = \frac{\text{Height}_{\text{tower}}}{\text{Shadow}_{\text{tower}}} \Rightarrow \frac{6}{4} = \frac{\text{Height}_{\text{tower}}}{28}$$

$$\text{Height}_{\text{tower}} = \frac{6}{4} \times 28 = 6 \times 7 = 42 \text{ m}$$

33. $\triangle ABC \sim \triangle DEF$ and their areas be respectively 64 cm^2 and 121 cm^2 . If $EF = 15.4$ cm, the value of BC is.

- (a) 11.2 cm (b) 15.4 cm (c) 6.4 cm (d) None of these

Ans: (a) 11.2 cm

$$\frac{\text{Area}(\triangle ABC)}{\text{Area}(\triangle DEF)} = \left(\frac{BC}{EF}\right)^2 \Rightarrow \frac{64}{121} = \left(\frac{BC}{15.4}\right)^2$$

$$\sqrt{\frac{64}{121}} = \frac{BC}{15.4} \Rightarrow \frac{8}{11} = \frac{BC}{15.4} \Rightarrow BC = \frac{8}{11} \times 15.4 = 8 \times 1.4 = 11.2 \text{ cm}$$

34. ABC and BDE are two equilateral triangles such that D is the midpoint of BC . Ratio of the areas of triangles ABC and BDE is

- (a) 2 : 1 (b) 1 : 2 (c) 4 : 1 (d) 1 : 4

Ans: (c) 4: 1

35. Areas of two similar triangles are in the ratio 4: 9. Sides of these triangles are in the ratio

- (a) 2 : 3 (b) 4 : 9 (c) 81 : 16 (d) 16 : 81

Ans: (a) 2: 3

36. A vertical pole of length 20 m casts a shadow 10 m long on the ground and at the same time a tower casts a shadow 50 m long, then the height of the tower.

- (a) 100 m (b) 120 m (c) 25 m (d) None of these

Ans: (a) 100 m

$$\frac{\text{Height}_{\text{pole}}}{\text{Shadow}_{\text{pole}}} = \frac{\text{Height}_{\text{tower}}}{\text{Shadow}_{\text{tower}}} \quad \frac{20}{10} = \frac{\text{Height}_{\text{tower}}}{50}$$

$$\text{Height}_{\text{tower}} = 2 \times 50 = 100 \text{ m}$$

37. The areas of two similar triangles are in the ratio 16: 9. The corresponding sides of these triangles are in the ratio

- (a) 2 : 3 (b) 4 : 3 (c) 81 : 16 (d) 16 : 81

Ans: (b) 4: 3

38. The areas of two similar triangles are in respectively 9 cm^2 and 16 cm^2 . The ratio of their corresponding sides is

- (a) 2 : 3 (b) 3 : 4 (c) 4 : 3 (d) 4 : 5

Ans: (b) 3: 4

39. Two isosceles triangles have equal angles and their areas are in the ratio 16: 25. The ratio of their corresponding heights is

- (a) 3 : 2 (b) 5 : 4 (c) 5 : 7 (d) 4 : 5

Ans: (d) 4: 5

40. If $\triangle ABC$ and $\triangle DEF$ are similar such that $2AB = DE$ and $BC = 8 \text{ cm}$, then $EF =$

- (a) 16 cm (b) 112 cm (c) 8 cm (d) 4 cm

Ans: (a) 16 cm

$$\frac{AB}{DE} = \frac{BC}{EF} \Rightarrow \frac{1}{2} = \frac{8}{EF} \Rightarrow EF = 2 \times 8 = 16 \text{ cm.}$$

41. XY is drawn parallel to the base BC of a $\triangle ABC$ cutting AB at X and AC at Y. If $AB = 4BX$ and $YC = 2 \text{ cm}$, then $AY =$

- (a) 2 cm (b) 6 cm (c) 8 cm (d) 4 cm

Ans: (b) 6 cm

$$\frac{AX}{XB} = \frac{3BX}{BX} = 3 \Rightarrow \frac{AY}{YC} = 3 \Rightarrow \frac{AY}{2} = 3 \Rightarrow AY = 3 \times 2 = 6$$

42. Two poles of height 6 m and 11 m stand vertically upright on a plane ground. If the distance between their foot is 12 m, the distance between their tops is

- (a) 14 cm (b) 12 cm (c) 13 cm (d) 11 cm

Ans: (c) 13 cm

43. If D, E, F are midpoints of sides BC, CA and AB respectively of $\triangle ABC$, then the ratio of the areas of triangles DEF and ABC is

- (a) 2 : 3 (b) 1 : 4 (c) 1 : 2 (d) 4 : 5

Ans: (b) 1: 4

44. If $\triangle ABC$ and $\triangle DEF$ are two triangles such that $AB / DE = BC / EF = CA / FD = 2 / 5$, then $ar(\triangle ABC) / ar(\triangle DEF) =$
 (a) 2: 5 (b) 4: 25 (c) 4: 15 (d) 8: 125

Ans: (b) 4: 25

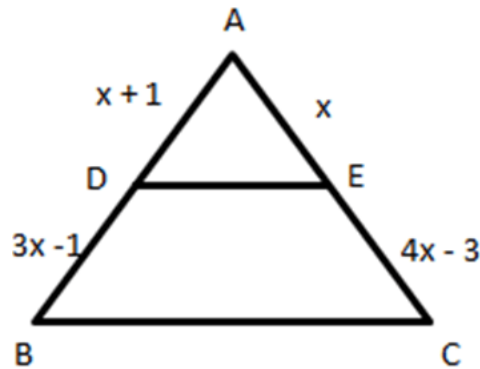
45. In triangles ABC and DEF , $\angle A = \angle E = 40^\circ$, $AB : ED = AC : EF$ and $\angle F = 65^\circ$, then $\angle B =$
 (a) 35° (b) 65° (c) 75° (d) 85°

Ans: (c) 75°

46. If ABC and DEF are similar triangles such that $\angle A = 47^\circ$ and $\angle E = 83^\circ$, then $\angle C =$
 (a) 50° (b) 60° (c) 70° (d) 80°

Ans: (a) 50°

47. What is the value of x ?



- (a) 0 (b) 1 (c) 2 (d) 3

Ans: (b) 1

$\therefore AD/DB = AE/EC$ by Thales Theorem,
 $x+1/3x-1 = x/4x-3 \Rightarrow x = -3, 1$
 Since lengths cannot be negative so $x = 1$

48. The perimeters of two similar triangles ABC , PQR is 64 cm and 24 cm respectively. If PQ is 12 cm what will be the length of AB ?
 (a) 30 cm (b) 32 cm (c) 12 cm (d) 16 cm

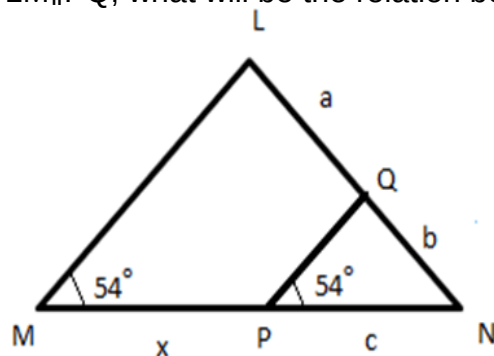
Ans: (b) 32 cm

$\frac{\text{Perimeter of } \triangle ABC}{\text{Perimeter of } \triangle PQR} = \frac{AB}{PQ}$
 $\frac{64}{24} = \frac{AB}{12} \Rightarrow AB = \frac{64 \times 12}{24} = 32 \text{ cm}$

49. D and E are the midpoints of side AB and AC of a triangle ABC , respectively and $BC = 6$ cm. If $DE \parallel BC$, then the length (in cm) of DE is:
 (a) 2.5 cm (b) 3 cm (c) 5 cm (d) 6 cm

Ans: (b) 3 cm

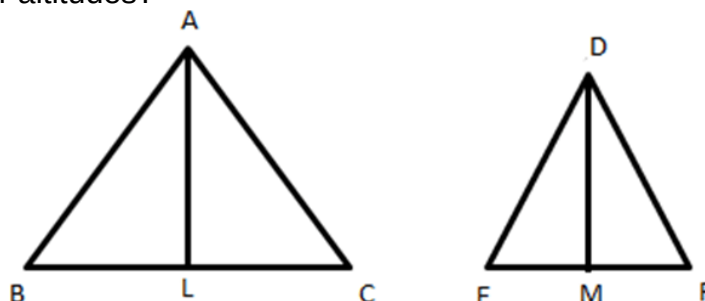
50. In the given figure, $LM \parallel PQ$, what will be the relation between x , a , b and c ?



- (a) $a = c b$ (b) $ab = cx$ (c) $b x = ac$ (d) $c b = a x$

Ans: (c) $b x = ac$

51. In two similar triangles $\triangle ABC$ and $\triangle DEF$, $AB = 15\text{cm}$, $DE = 5\text{cm}$. If AL and DM are the altitudes of the triangles ABC , DEF respectively, then what will be the ratio of their altitudes?



- (a) $3 : 1$ (b) $1 : 3$ (c) $1 : 2$ (d) $2 : 1$

Ans: (a) $3 : 1$

52. If $\angle A = \angle P$, $\angle B = \angle Q$, $\angle C = \angle R$ then, $\triangle ABC$ & $\triangle PQR$ are similar according to which test?

- (a) AAA test (b) AA test (c) SAS test (d) SSS test

Ans: (a) AAA test

53. If $\angle D = \angle L$, $\angle E = \angle M$ then, $\triangle DEF$ & $\triangle LMN$ are similar according to which test?

- (a) AAA test (b) AA test (c) SAS test (d) SSS test

Ans: (b) AA test

54. If $ABXY = BCYZ = ACXZ$ then, $\triangle ABC$ & $\triangle XYZ$ are similar according to which test?

- (a) AAA test (b) AA test (c) SAS test (d) SSS test

Ans: (d) SSS test

55. The altitude of a right triangle is 7 cm less than its base. If the hypotenuse is 13 cm, the other two sides of the triangle are equal to:

- (a) Base = 10cm and Altitude = 5cm (b) Base = 12cm and Altitude = 5cm
(b) Base = 14cm and Altitude = 10cm (d) Base = 12cm and Altitude = 10cm

Ans: (b) Base = 12cm and Altitude = 5cm

56. A circle has a number of tangents equal to

- (a) 0 (b) 1 (c) 2 (d) Infinite

Ans: (d) Infinite

57. A tangent intersects the circle at:

- (a) One point (b) Two distinct point (c) At the circle (d) None of these

Ans: (a) One point

58. A circle can have _____ parallel tangents at a single time.

- (a) One (b) Two (c) Three (d) Four

Ans: (b) Two

59. If the angle between two radii of a circle is 110° , then the angle between the tangents at the ends of the radii is:

- (a) 90° (b) 50° (c) 70° (d) 40°

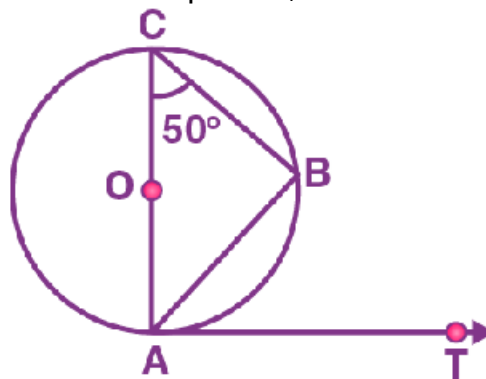
Ans: (c) 70°

60. The length of the tangent from an external point A on a circle with centre O is

- (a) Always greater than OA (b) Equal to OA
(b) Always less than OA (d) Cannot be estimated

Ans: (c) Always less than OA

61. AB is a chord of the circle and AOC is its diameter such that $\angle ACB = 50^\circ$. If AT is the tangent to the circle at the point A, then $\angle BAT$ is equal to

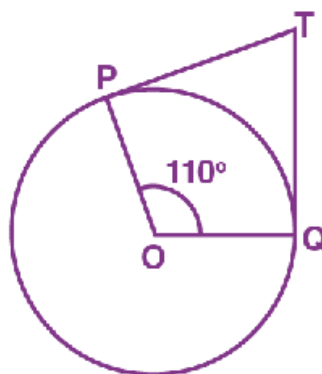


- (a) 65° (b) 60° (c) 50° (d) 40°

Ans: (c) 50°

$$\angle A = 180^\circ - (90^\circ + 50^\circ) \Rightarrow \angle A = 40^\circ \Rightarrow \angle BAT = 90^\circ - 40^\circ = 50^\circ$$

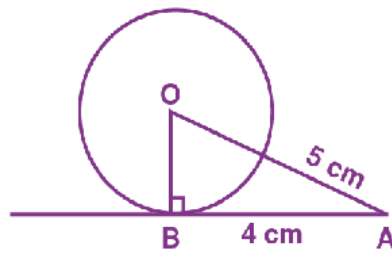
62. If TP and TQ are the two tangents to a circle with centre O so that $\angle POQ = 110^\circ$, then $\angle PTQ$ is equal to



- (a) 60° (b) 70° (c) 80° (d) 90°

Ans: (b) 70°

63. The length of a tangent from a point A at a distance 5 cm from the centre of the circle is 4 cm. The radius of the circle is:



- (a) 3 cm (b) 5 cm (c) 7 cm (d) 10 cm

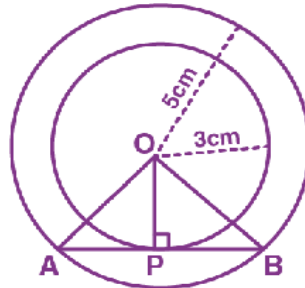
Ans: (a) 3 cm

64. If a parallelogram circumscribes a circle, then it is a:

- (a) Square (b) Rectangle (c) Rhombus (d) None of these

Ans: (c) Rhombus

65. Two concentric circles are of radii 5 cm and 3 cm. The length of the chord of the larger circle which touches the smaller circle is:



- (a) 8 cm (b) 10 cm (c) 12 cm (d) 18 cm

Ans: (a) 8 cm

66. If angle between two radii of a circle is 130° , the angle between the tangents at the ends of the radii is

- (a) 90° (b) 50° (c) 70° (d) 40°

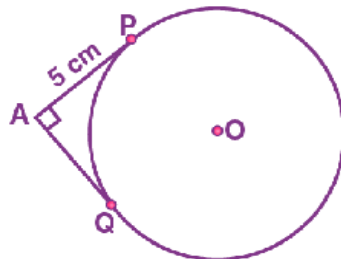
Ans: (b) 50°

67. A line intersecting a circle in two points is called a _____.

- (a) Secant (b) Chord (c) Diameter (d) Tangent

Ans: (a) Secant

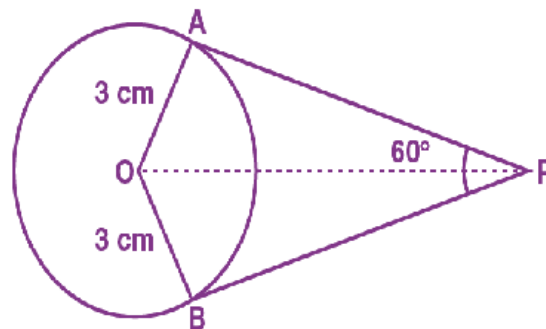
68. In the figure below, the pair of tangents AP and AQ drawn from an external point A to a circle with centre O are perpendicular to each other and length of each tangent is 5 cm. Then the radius of the circle is



- (a) 10 cm (b) 7.5 cm (c) 5 cm (d) 2.5 cm

Ans: (c) 5 cm

69. If two tangents inclined at an angle 60° are drawn to a circle of radius 3 cm, then length of each tangent is equal to



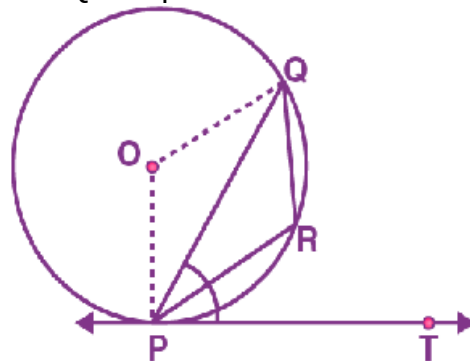
- (a) $(3/2)\sqrt{3}$ cm (b) 6 cm (c) 3 cm (d) $3\sqrt{3}$ cm
Ans: (d) $3\sqrt{3}$ cm

70. The tangent to a circle is _____ to the radius through the point of contact.

- (a) Parallel (b) Perpendicular (c) Perpendicular bisector (d) Bisector

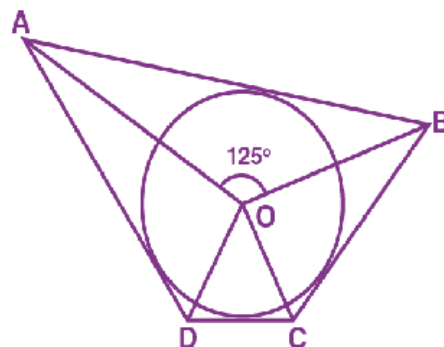
Ans: (b) Perpendicular

71. In the figure below, PQ is a chord of a circle and PT is the tangent at P such that $\angle QPT = 60^\circ$. Then $\angle PRQ$ is equal to



- (a) 135° (b) 150° (c) 120° (d) 110°
Ans: (c) 120°

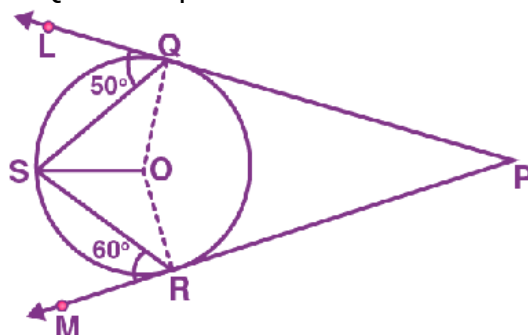
72. In the below figure, if $\angle AOB = 125^\circ$, then $\angle COD$ is equal to



- (a) 62.5° (b) 45° (c) 35° (d) 55°
Ans: (d) 55°

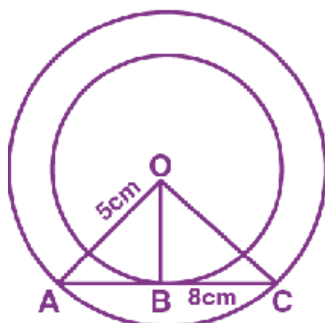
$\angle COD = 180^\circ - 125^\circ = 55^\circ$

73. In the figure, PQL and PRM are tangents to the circle with centre O at the points Q and R, respectively and S is a point on the circle such that $\angle SQL = 50^\circ$ and $\angle SRM = 60^\circ$. Then $\angle QSR$ is equal to



- (a) 40° (b) 60° (c) 70° (d) 80°
Ans: (c) 70°

74. Out of the two concentric circles, the radius of the outer circle is 5 cm and the chord AC of length 8 cm is a tangent to the inner circle. The radius of the inner circle will be



- (a) 3 cm (b) 4 cm (c) 2.5 cm (d) 2 cm
Ans: (a) 3 cm

75. The distance between two parallel tangents of a circle is 18 cm, then the radius of the circle is

- (a) 8 cm (b) 10 cm (c) 9 cm (d) 7.5 cm
Ans: (c) 9 cm

76. If at some time, the length of the shadow of a tower is $\sqrt{3}$ times its height, then the angle of elevation of the sun, at that time is:

- (a) 15° (b) 30° (c) 45° (d) 60°
Ans: (b) 30°

77. At some time of the day, the length of the shadow of a tower is equal to its height. Then, the sun's altitude at that time is:

- (a) 30° (b) 60° (c) 90° (d) 45°
Ans: (d) 45°

78. A person is flying a kite at a height of 30 m from the horizontal level. The length of string from the kite to the person is 60 m. Assuming that there is no slack in the string, the angle of elevation of kite to the horizontal level is:

- (a) 45° (b) 30° (c) 60° (d) 90°
Ans: (b) 30°

79. The angle of depression of a car, standing on the ground, from the top of a 75 m high tower is 30° . The distance of the car from the base of tower (in m) is:

- (a) $25\sqrt{3}$ (b) $50\sqrt{3}$ (c) $75\sqrt{3}$ (d) 150

Ans: (c) $75\sqrt{3}$

80. The angle of elevation of the top of a 15 m high tower at a point 15 m away from the base of tower is:

- (a) 30° (b) 60° (c) 45° (d) 75°

Ans: (c) 45°

81. A man standing at a height 6 m observes the top of a tower and the foot of tower at angles of 45° and 30° of elevation and depression respectively. The height of tower is:

- (a) $6\sqrt{3}$ m (b) 12 m (c) $6(\sqrt{3} - 1)$ (d) $6(\sqrt{3} + 1)$ m

Ans: (d) $6(\sqrt{3} + 1)$ m

82. Two poles are 25 m and 15 m high and the line joining their tops makes an angle of 45° with the horizontal. The distance between these poles is:

- (a) 5 m (b) 8 m (c) 9 m (d) 10 m

Ans: (d) 10 m

83. A 6 feet tall man finds that the angle of elevation of a 24 feet high pillar and the angle of depression of its base are complementary angles. The distance of man from the pillar is:

- (a) $4\sqrt{3}$ feet (b) $6\sqrt{3}$ feet (c) $8\sqrt{3}$ feet (d) $10\sqrt{3}$ feet

Ans: (b) $6\sqrt{3}$ feet

84. A lamp post $5\sqrt{3}$ m high casts a shadow 5 m long on the ground. The sun's elevation at this point is:

- (a) 30° (b) 45° (c) 60° (d) 90°

Ans: (c) 60°

85. The angle of elevation of the top of a tower from a point P on the ground is α . After walking a distance d towards the foot of the tower, angle of elevation is found to be β . Then

- (a) $\alpha < \beta$ (b) $\alpha > \beta$ (c) $\alpha = \beta$ (d) None of these

Ans: (a) $\alpha < \beta$

86. A ladder makes an angle of 60° with the ground, when placed along a wall. If the foot of ladder is 8 m away from the wall, the length of ladder is:

- (a) 4 m (b) 8 m (c) $8\sqrt{2}$ m (d) 16 m

Ans: (d) 16 m

87. If the height and length of a shadow of a man are the same, then the angle of elevation of sun is:

- (a) 30° (b) 60° (c) 45° (d) 15°

Ans: (c) 45°

88. The angle of elevation of the top of a tower from a point on the ground 30 m away from the foot is 30° . The height of the tower is:

- (a) 30 m (b) $10\sqrt{3}$ (c) 20 m (d) $10\sqrt{2}$ m

Ans: (b) $10\sqrt{3}$

89. To draw a pair of tangents to a circle which are inclined to each other at angle x° , it is required to draw tangents at the end points of those two radii of the circle, the angle between which is

- (a) $180^\circ - x^\circ$ (b) $90^\circ + x^\circ$ (c) $90^\circ - x^\circ$ (d) $180^\circ + x^\circ$

Ans: (a) $180^\circ - x^\circ$

90. PT and PS are tangents drawn to a circle, with centre C, from a point P. If $\angle TPS = 50^\circ$, then the measure of $\angle TCS$ is

- (a) 150° (b) 130° (c) 120° (d) 100°

Ans: (b) 130°

91. If two tangents are drawn at the end points of two radii of a circle which are inclined at 120° to each other, then the pair of tangents will be inclined to each other at an angle of

- (a) 60° (b) 90° (c) 100° (d) 120°

Ans: (a) 60°

92. A point O is at a distance of 10 cm from the centre of a circle of radius 6 cm. How many tangents can be drawn from point O to the circle?

- (a) 2 (b) 1 (c) Infinite (d) 0

Ans: (a) 2

93. In division of a line segment AB, any ray AX making angle with AB is

- (a) Right angle (b) Obtuse angle
(c) Any arbitrary angle (d) Acute angle

Ans: (d) Acute angle

94. Which theorem criterion we are using in giving the justification of the division of a line segment by usual method?

- (a) SSS criterion (b) Area theorem (c) BPT (d) Pythagoras theorem

Ans: (c) BPT

95. The tops of two poles of height 20 m and 14 m are connected by a wire. If the wire makes an angle of 30° with horizontal, the length of the wire is

- (a) 6 m (b) 10 m (c) 12 m (d) 20 m

Ans: (c) 12 m

96. If the length of the shadow of a tower is increasing, then the angle of elevation of the sun is

- (a) Also increasing (b) Decreasing
(c) Remains unaffected (d) Don't have any relation with length of shadow

Ans: (b) Decreasing

97. A ladder 15 metres long just reaches the top of a vertical wall. If the ladder makes an angle of 60° with the wall, then the height of the wall will be

- (a) 7.5m (b) 7.7m (c) 8.5m (d) 8.8m

Ans: (a) 7.5m

98. An observer 1.5 metres tall is 20.5 metres away from a tower 22 metres high. Determine the angle of elevation of the top of the tower from the eye of the observer.

- (a) 30° (b) 45° (c) 60° (d) 90°

Ans: (b) 45°

99. The angles of elevation of the top of a tower from two points' distant s and t from its foot are complementary. Then the height of the tower is:

- (a) $s t$ (b) $s^2 t^2$ (c) $\sqrt{s t}$ (d) s / t

Ans: (c) $\sqrt{s t}$

100. The shadow of a tower standing on a level plane is found to be 50 m longer when Sun's elevation is 30° than when it is 60° . Then the height of tower is:

- (a) $20\sqrt{3}$ (b) $25\sqrt{3}$ (c) $10\sqrt{3}$ (d) $30\sqrt{3}$

Ans: (b) $25\sqrt{3}$